



Knowledge tensions and decision-making challenges in open innovation: Standardization as a de-biasing mechanism

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ABSTRACT

Complexities in external knowledge evaluation present significant “limits” to open innovation (OI), challenging decision making in the processes of searching, accessing, and using external knowledge for recombination, which are central to the OI paradigm. Grounded in the theory of attitudes, this study investigates the role of standardization in preventing the negative behavioral influence of “not invented here” (NIH) and “not-shared-here” (NSH) attitudes on knowledge search and sourcing decisions by debiasing decisional paths associated with reluctance to adopt OI. Using hurdle model estimations on 600 small-and medium-sized enterprises, the results of the study reveal that NIH and NSH attitudes are the origin of decision-making biases at different stages of OI implementation, where external knowledge is the focus of evaluation. Proactive standardization efforts are shown to be effective in countering the impact of NIH and NSH attitudes on knowledge sourcing decisions and on the intensity of knowledge sourcing, typically leading to bounded search, inward decisions, and anticipated termination of projects. This study provides insights into the influence of standardization efforts specifically addressing challenges related to the evaluation of external knowledge, where the likelihood of failure is higher and associated costs are significant.

1. Introduction

The dynamic process through which firms adopt and implement Open Innovation (OI) crucially involves organizational efforts supporting the purposive search, access, and use of relevant external knowledge for recombination (Chesbrough and Bogers, 2014; Chesbrough, 2019). From a process perspective, the process from closed to open innovation is a path of inter-organizational learning (Chiaroni et al., 2010; Seo and Park, 2022), where these different stages (sub-processes) involve decisions based on an evaluation and interpretation of external knowledge and its potential use through recombination with internal knowledge (Antons and Piller, 2015; Hannen et al., 2019). As firms confront an increasing variety of external knowledge sources, effective engagement in OI necessitates reshaping their learning processes (Hodgkinson and Healey, 2011) through the development of methods, structures, and systems that act as interfaces for knowledge exchange (Chiaroni et al., 2011; Kim and Ahn, 2020; Van Der Meer, 2007).

At the same time, increasing openness to external knowledge also incurs important trade-offs and paradoxical tensions (Dabić et al., 2023; Ritala and Stefan, 2021), often originating at the level of individuals

(Antons and Piller, 2015; Ritala and Stefan, 2021) and producing substantial negative effects on decision making at the organizational level (Hannen et al., 2019; Obradović et al., 2021). The literature offers several explanations for why organizations, especially smaller ones, face management challenges in evaluating and using external knowledge internally (Kim and Ahn, 2020). Boundaries to external search strategies are often imposed as accessing and using external knowledge requires simultaneous investments in R&D, skills, and competencies for its evaluation (Martini et al., 2017), raising search and absorption costs (Audretsch and Belitski, 2020; Laursen and Salter, 2006). Here, the benefits of establishing relationships with external actors to enlarge the pool of relevant knowledge resources must be balanced with increasing complexity in external knowledge evaluation and interpretation, diverting resources from R&D, and limiting the development of absorptive capacity (Ahn et al., 2017; Marullo et al., 2020). Furthermore, sourcing decisions are significantly challenged as the intensity of knowledge sharing increases (Audretsch and Belitski, 2023; Kim and Ahn, 2020), as individuals may become increasingly reluctant to disclose sensitive information about untested ideas or provide insight into internal knowledge and skills to other firms (Burcharth et al., 2014;

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Veugelers, 1997).

Since Katz and Allen (1982), “not invented here” (NIH) and “not shared here” (NSH) attitudes have been identified as prominent individual-level psychological barriers toward external knowledge access and utilization, with negative consequences on decision making in OI (Agrawal et al., 2010; Burcharth et al., 2014; Grosse Kathoefer and Leker, 2012). While NIH attitudes refer to negative dispositions of individuals toward the internal use of knowledge that is perceived as “external” to the group or the organization (Antons and Piller, 2015), NSH attitudes stem from an unfavorable perception of the outside application of knowledge developed inside the firm through sourcing and recombination (Burcharth et al., 2014). NIH and NSH reflect a misalignment of incentives and diverging interests in OI (West et al., 2014), with clear cascading effects in the sub-processes of searching, accessing, and using external knowledge, and consequent adverse effects on decision-making. The typical behavioral responses caused by NIH and NSH attitudes are underutilization or rejection of external knowledge and a substantial reluctance to organize knowledge management tasks externally (Antons et al., 2017; Burcharth et al., 2014; Lichtenthaler, 2011) due to incorrect evaluations and judgment errors (Antons and Piller, 2015). In the logic of micro foundations (Felin et al., 2015), NIH and NSH attitudes therefore represent individual-level factors that may hinder achieving collective, organizational-level outcomes in OI and can be regarded as early, individual-level sources of knowledge tensions (Dabić et al., 2023; Stefan et al., 2022).

While previous research has explored various organizational measures to contain NIH and NSH (e.g., Grosse Kathoefer and Leker, 2012; Katz and Allen, 1982) or to limit their formation through the development of routines (e.g., Herzog and Leker, 2010; Kim and Ahn, 2020), less attention has been paid to indirect countermeasures (Hannen et al., 2019), seeking to “prevent the attitude from influencing behavior instead of directly aiming at changing the attitude” (Antons and Piller, 2015, p. 207). Drawing on debiasing principles from attitudes theory (Fabrigar et al., 2005; Soll et al., 2015), and in response to recent calls for research bridging concepts and methods from psychology and cognitive science in understanding the limits of OI (Dabić et al., 2023; Hannen et al., 2019), this study proposes knowledge standardization as an indirect NIH and NSH countermeasure in OI adoption. Specifically, this study explores the mechanisms through which standardization, primarily regarded as a market diffusion instrument or a regulatory tool, can also contain the negative consequences of individuals’ reliance on NIH and NSH attitudes as information processing heuristics on external knowledge search and sourcing decisions. By facilitating the evaluation, interpretation, sharing, and integration of knowledge, standardization seeks to debias external knowledge search and sourcing behavior in the process from closed to open innovation (Hannen et al., 2019; Soll et al., 2015).

A conceptual framework integrating the literature streams on standardization, knowledge search, and OI is proposed to examine whether and how the pursuit of standardization—“the codification of knowledge elements of an industry’s technology or, simply, information that is relevant to further innovation” (Tassey, 2000, p. 588)—is effective in countering the impact of emerging NIH and NSH attitudes on decision making in different stages of OI adoption.

For empirical evidence, a double hurdle estimation procedure was used to model OI implementation from a process perspective, where different stages (subprocesses) involve the selection, access, and use of external knowledge for recombination. The analysis results suggest that the pursuit of standards harnesses the efficacy of debiasing external search strategies by counteracting the negative influence of NIH attitudes on knowledge sourcing decisions. Aligned with their economic function, standards act as collective constructs that facilitate the evaluation of external knowledge and reduce uncertainty, enabling its application in different contexts (Blind, 2017; Blind and Gauch, 2009). Furthermore, we find that organizational efforts toward internal knowledge standardization (such as, the collaborative development of

product specifications) partially offset the negative impact of NSH on the depth of sourcing efforts, facilitating knowledge recombination aimed at furthering innovation through the formation of collective consensus (Blind and Mangelsdorf, 2016; Bogers and Horst, 2014; Xie et al., 2016). Our study extends the current understanding of the micro foundations of OI challenges (Bogers et al., 2017; Dabić et al., 2023; Stefan et al., 2022) by investigating the influence of knowledge standardization on firms’ search pathways in OI (Xie et al., 2016), specifically addressing decision-making challenges related to the evaluation of external knowledge, where the risks of failure are typically higher (Audretsch and Belitski, 2020; West and Bogers, 2013).

In the next section, we introduce and discuss our conceptual framework, from which the hypotheses are derived. This is followed by a description of the research design, the data, and the empirical strategy. Subsequent sections present the analytical results and discuss the main implications, including actionable insights for practitioners and policy makers.

2. Theoretical background

In its original conceptualization, the OI paradigm was defined as an open R&D process that combines internal and external knowledge to create value and overcome the constraints imposed by internal knowledge (Chesbrough, 2003, 2006). Further theoretical developments have framed the transition from closed to OI settings as a dynamic decision-making process in which firms choose between conducting internal innovation activities or accessing, harnessing, and recombining external knowledge at different stages (Dahlander and Gann, 2010; Huizingh, 2011). In this context, the adoption and implementation of OI can be seen as an evolutionary path in which firms engage in learning through searching, selecting, creating, and adapting to novelty (Hölzl, 2005; Koput, 1997).

Recent research has demonstrated an increasing interest in exploring knowledge-related tensions in OI. The discussion about challenges in implementing OI (Dahlander and Gann, 2010) has recently evolved toward a more refined understanding of its “limits” (Dabić et al., 2023), evident in decreasing returns in knowledge sourcing (Laursen and Salter, 2006), evaluation problems (Audretsch and Belitski, 2023), and the emergence of paradoxical tensions related to knowledge sharing for recombination (Laursen and Salter, 2014; Ritala and Stefan, 2021).

As the OI literature matures, there is still much to understand regarding the contingencies or boundary conditions of its adoption, particularly in the context of knowledge search and sourcing decisions (Dabić et al., 2023). This involves an in-depth examination of the prominent trade-offs and incentive problems intrinsic to OI implementation (Audretsch and Belitski, 2023; West et al., 2014). Tensions arise between the benefits of external knowledge acquisition and assimilation - i.e., the development of potential absorptive capacity (Enkel and Heil, 2014)- and the degree of external knowledge heterogeneity, complexity, and specialization (Koput, 1997). Searching and sourcing knowledge that originates from different disciplinary fields, other organizations, or geographically distant sources adds complexity to the evaluation process because of the heterogeneity of information that must be reviewed and evaluated, distracting resources and attention from internal R&D activities and potentially introducing constraints, negative feedback, or measurement errors (Antons and Piller, 2015; Chesbrough, 2004).

Boundary conditions to knowledge sourcing are imposed to avoid increasing search costs and absorptive capacity problems, with significant consequences, including anticipated termination of projects (Purdy et al., 2023), inward decisions (West et al., 2014) and bounded search approaches (Audretsch and Belitski, 2020). Additionally, the opportunities of sharing knowledge for recombination (Cassiman and Veugelers, 2006) come with challenges arising from unintended leakage or misappropriation of that knowledge (Heiman and Nickerson, 2004; Laursen and Salter, 2014; West, 2006). Knowledge disclosure becomes a

central issue when specialized knowledge or sensitive information needs to be revealed and leveraged through its application (Coff et al., 2006; Ritala et al., 2015). Thus, knowledge search and sourcing decisions in OI paradoxically become challenging as the “openness” to external knowledge increases (Laursen and Salter, 2006; Ritala and Stefan, 2021; West and Gallagher, 2006).

2.1. NIH, NSH, and decision-making challenges in OI

Knowledge search and sourcing decisions are major strategic issues in OI implementation. However, these activities, which are crucial for accessing and using relevant external knowledge, often cause paradoxical tensions (Ritala and Stefan, 2021). Psychological and cultural factors, particularly attitudes of individuals toward external knowledge (Katz and Allen, 1982), and their responses to emerging tensions (Stefan et al., 2022), are among the individual-level attributes that most significantly impact decision making at the organizational level. As attitudes represent crucial elements in balancing the processes of external interactions and knowledge exchange (Ajzen, 2001; Bohner and Dickel, 2011), attitudes also act as sources of biased decision making as firms progressively embrace OI (Antons et al., 2017; Antons and Piller, 2015).

The “NIH and NSH syndromes”, conceptualized as negative dispositions toward acquiring external knowledge and using internal knowledge for recombination, have been recognized as among the most challenging individual-level barriers to the adoption of OI, often making managerial efforts insufficient to achieve objectives (Chesbrough and Crowther, 2006; Chesbrough et al., 2006). Such attitudes generate misalignment of incentives for generating knowledge spillovers in OI (West et al., 2014; West and Gallagher, 2006), with negative implications for the adoption of OI (Burcharth et al., 2014). As such, NIH and NSH attitudes can be considered crucial attributes at the individual level within organizations, having cascading effects at the higher (organizational) level in OI implementation (Hannen et al., 2019). From a multi-level perspective, understanding where and how the behavioral consequences of such effects manifest is crucial for advancing the literature on the micro foundations of OI challenges.

The NIH attitude — a negative disposition toward knowledge that “has to cross a disciplinary, spatial, or organizational boundary” (Antons et al., 2017, p. 1227) has its origin in the “tendency of a stable group to believe it possesses a monopoly of knowledge in its field” (Katz and Allen, 1982, p. 7). NIH therefore differs from a reasonable rejection of ideas based on objective evidence. Instead of rationally evaluating new knowledge, NIH attitudes lead to a systematic underestimation of its value based on fact that it is perceived as “external” to the group or the organization in which individuals operate. As noted by Antons and Piller (2015), the fundamental sources of NIH are knowledge externalities, caused by the degree of knowledge heterogeneity in terms of functional, geographic, and disciplinary boundaries — along with its complexity and specialization. Thus, the intensity of NIH attitudes mirrors the extent to which a firm’s internal knowledge base differs from other types of external knowledge based on operational function, geographical, or cultural context.

In OI adoption, NIH is reflected in decision-making biases of growing relevance as firms progressively open up to external knowledge (Antons and Piller, 2015; Kim and Ahn, 2020). The process of searching for external knowledge inputs through which firms initiate OI often involves “trial and error” approaches to “understand the norms and routines of different external knowledge channels” (Laursen and Salter, 2006, p. 135). In this process, decision making is influenced by considerable uncertainty, and time and managerial attention are required to evaluate the potential of different external sources and to understand which source(s) of knowledge will be the most relevant to maximize the return on future external collaborations and internal R&D investments (Audretsch and Belitski, 2020).

As firms are inevitably confronted with an increase in complexity due to the heterogeneity of external knowledge and uncertainty in its

evaluation, internal resistance (due, e.g., to past unrewarding experiences) may lead to judgment errors in the evaluation of external knowledge and eventually to the rejection of opportunities for enhancing the internal knowledge base (Levinthal and March 1993). There are numerous reasons that trigger the NIH attitude as an internal cultural barrier to external knowledge sourcing decisions. Protective attitudes lead to overestimation of internally generated knowledge when external knowledge is viewed as a threat to internal R&D projects and employee capabilities (Gupta and Govindarajan, 2000). Additionally, NIH attitudes are a source of confirmation bias when firms process information selectively and filter out external ideas that do not conform to their internal culture or pre-existing beliefs, leading to the rejection of potentially useful knowledge that is considered “unfamiliar” (Antons and Piller, 2015). Confirmation biases often result in redundant knowledge search, as firms fail to look beyond what they “expect” to find (Ajzen, 2001), thus impeding sourcing decisions.

As the of heterogeneity and complexity of (new) external knowledge plays a vital role in escalating NIH attitudes, sourcing decisions will be affected by higher NIH attitudes with a higher probability of rejecting knowledge in cases where it would actually be valuable (Antons and Piller, 2015; Kobarg et al., 2019). In this case, higher NIH attitudes will exert tensions (Bohner and Dickel, 2011), acting as a deterrent to external knowledge sourcing. This leads to the following hypothesis:

H1. *Higher NIH attitudes negatively affect knowledge sourcing decisions in OI.*

As an extension of NIH, NSH has been conceptualized as a negative disposition toward the outside application of internal knowledge, resulting in deliberate barriers to the generation of knowledge spillovers in OI (Burcharth et al., 2014; West et al., 2014). In contrast to NIH attitudes, which generate biases in the process of evaluation of external knowledge and impact subsequent sourcing decisions, NSH attitudes may escalate when the intensity of interaction with external sources increases and knowledge flows become bidirectional. Lowering the incentives to disclose sensitive information to partners to maximize appropriability, NSH limits the depth of external knowledge sourcing efforts and results in the rejection of external interactions when organizing knowledge management tasks (Lichtenthaler and Ernst, 2006). Thus, the impact of NIH and NSH on the adoption of OI can be associated with the two sub-processes of external knowledge sourcing and use for recombination in OI, as search and sourcing decisions lead to higher dependency on the interaction between internal and external knowledge (Ahn et al., 2017b; Kim and Ahn, 2020). While the evolution from exploratory search to external knowledge sourcing involves challenges in knowledge evaluation (Cruz-González et al., 2015; Laursen and Salter, 2006), the intensity of knowledge sourcing efforts, demanding a higher degree of “openness” to specific channels, may trigger protective employee attitudes that minimize the risk of unintended disclosure of ideas, knowledge, or core competencies (Audretsch and Belitski, 2020; Kobarg et al., 2019). Hence, the perception of a potential “sell-out” of internal knowledge in presence of NSH would lead to a misalignment of incentives, limiting the coordination of knowledge sourcing efforts (Coff et al., 2006; Grant, 1996) and resulting in inward decisions (Antons et al., 2017).

H2. *Higher NSH attitudes negatively affect the depth of knowledge sourcing efforts in OI.*

2.2. Debiasing external knowledge acquisition: the role of standardization in OI

2.2.1. Countermeasures to NIH and NSH attitudes

A growing body of empirical research has explored countermeasures to NIH and NSH attitudes, suggesting organizational mechanisms that directly address these negative attitudes to reduce them or prevent their formation (Hannen et al., 2019). Proposed organizational

countermeasures to NIH and NSH involve a reconfiguration of internal resources through both structural solutions, such as team rotation and employee integration in decision-making processes (Grosse Kathoefer and Leker, 2012; Katz and Allen, 1982), and contextual solutions focused on organizational climate (Barham et al., 2020; Herzog and Leker, 2010; Kim and Ahn, 2020). However, as implementing mechanisms aimed at changing employees' attitudes requires long-term effort and can be costly (Antons and Piller, 2015; Bohner and Dickel, 2011), their immediate effectiveness in improving OI project performance remains unproven (Natalicchio et al., 2018).

A more recent perspective, rooted in cognitive sciences, has shifted attention toward the effectiveness of indirect countermeasures to negative knowledge attitudes (Hannen et al., 2019). Instead of attempting to alter attitudes directly, indirect countermeasures aim to remove the distorting effects of attitudes on (i.e. at debiasing) decisions by influencing the attitude-behavior link (Ajzen, 2001; Soll et al., 2015). Previous studies have emphasized managing NSH and NIH by stimulating positive learning mechanisms (Barham et al., 2020; Herzog and Leker, 2010; Zollo and Winter 2002). These mechanisms target the sub-processes induced by NIH and NSH at the individual level and act on the mechanisms through which protective attitudes toward knowledge lead to negative preconceptions and biases that influence external knowledge search and sourcing decisions (Ahn et al., 2017; Bogers et al., 2018; Burcharth et al., 2014). From this perspective, prior evidence suggests that competence-building initiatives, as well as procedures aimed at intensifying contacts with external knowledge providers, help mitigate the effects of NIH and NSH attitudes on decision making, positively influencing OI adoption (Burcharth et al., 2014; Gesing et al., 2015). However, the mechanisms through which such indirect countermeasures affect search and sourcing decisions require a more fine-grained understanding and a detailed empirical examination (Burcharth et al., 2014; Dabić et al., 2023; Hannen et al., 2019; West et al., 2014). Given that NIH and NSH attitudes comprise two components— psychological bias (i.e., individuals' reliance on such attitudes as information processing heuristics) and behavioral response (i.e., judgment and decision-making) (Antons and Piller, 2015) — indirect countermeasures would be effective if they can lead to knowledge search and sourcing decisions that consider alternative outcomes, even when negative attitudes toward external knowledge are present (Lar-rick, 2004).

2.2.2. Standardization as an indirect countermeasure

Standardization - the process of developing product and process specifications on a set of criteria, guidelines, or technology regulation activities (Blind and Mangelsdorf, 2016) -, and its resulting constructs—standards—promote compatibility, interoperability, and knowledge transfer among firms, users, interest groups, and governments, based on collective consensus (Xie et al., 2016, p. 69). The literature has extensively explored the economic functions of standards as technical infrastructures with public good content that enhance the efficiency of economic activity (Bénézech et al., 2001) and their significant impact on innovation diffusion (Tassey, 2000; Utterback, 1996). The voluntary use of consensus-based standards enhances coordination (Belleflamme, 2002) and significantly contributes to effective knowledge transfer in OI (Grøtnes, 2009). In accordance with their economic function in innovation, standards are tied to knowledge search processes as they represent inputs to collaborative innovation (Blind, 2017), facilitating the identification of relevant external knowledge and its evaluation, enabling application (Blind and Gauch, 2009).

As maintained by the knowledge-based theory of the firm (Grant, 1996), the more implicit and specialized the knowledge embodied in technologies, patents, know-how, or ideas, the slower and costlier its exploitation across industries and countries (Teece, 1998) due to market imperfections (Fosfuri, 2006). The emergence of NIH attitudes in OI has been specifically attributed to knowledge externalities (Antons and Piller, 2015; Katz and Allen, 1982), and their negative impacts on

knowledge search and sourcing decisions are reinforced by the complexity and heterogeneity of external knowledge itself (Katila and Ahuja, 2002; Kobarg et al., 2019; Laursen and Salter, 2006). As attitudes provide individuals with a knowledge processing structure (Antons and Piller, 2015), the more sensitive the attitude is to a specific knowledge domain (i.e. the higher the cognitive distance between internal and external knowledge domains), the higher will be individuals' resistance in evaluating knowledge from outside channels, favoring the positive evaluation of internal knowledge over external sources (West et al., 2014).

In the presence of NIH attitudes, using standards as codified knowledge inputs would substantially reduce the complexity of its evaluation (Chesbrough, 2004; Koput, 1997), lowering the possibility of negative feedback or measurement errors due to confirmation biases, as individuals rely on NIH attitudes as information processing heuristics (Antons and Piller, 2015). By allowing a shared understanding of the types of knowledge that will be most relevant, standards aid evaluation processes, helping to focus on specific aspects of the innovation problem (Pierce et al., 2001). The use of standards for external knowledge sourcing has proven effective in firms' managerial practices as strategic communication tools to engage with potential external knowledge contributors. From the perspective of smaller firms, typically disposing of limited resources for the development of absorptive capacity, standards can prove essential for developing focused search strategies, enhancing internal capabilities, and aligning with external partners (Liao et al., 2003). In the context of small- and medium-sized enterprises (SMEs), the highest obstacles to external knowledge sourcing due to NIH attitudes occur when sourcing involves technological knowledge from R&D-intensive industries, and tend to increase where "openness" implies a high breadth of collaboration partners (Ahn et al., 2017b; Audretsch and Belitski, 2023; Kim and Ahn, 2020). In contrast to their larger counterparts, due to their slack organizational structure, SMEs cannot implement costly and long-term structural approaches (such as, team rotation among business units) aimed at preventing the development of NIH.

In this context, standardized knowledge can act as a mechanism supporting knowledge access (Nonaka et al., 1996) and de-biasing its acquisition (Hannen et al., 2019). Using standards in external knowledge search processes may increase the chance of vertical collaboration along the value chain and co-specialization with external partners. Easier access to external knowledge may further enhance repetitive collaboration, thereby strengthening the intensity of knowledge sourcing and absorption. Hence, in the transition from a broad search for external knowledge inputs to more intensive knowledge sourcing processes in OI, the use of standards would increase the likelihood that a firm will actively access external knowledge by counteracting the pressure exerted by NIH attitudes toward external knowledge rejection.

This leads to the third hypothesis.

H3. *The use of standards in knowledge search processes will counteract the negative impact of NIH attitudes on sourcing decisions in OI.*

Recent research has explored firms' independent efforts in standardization as a means of codifying knowledge to enhance collaborative innovation outcomes (Blind and Mangelsdorf, 2016; Xie et al., 2016). Standardization initiatives led by individual firms can be seen as deliberate attempts to develop codified knowledge inputs for their products, processes, and related innovations (Xie et al., 2016, p. 71). As routinized procedures for codifying technological knowledge into specification documents, efforts toward knowledge standardization enable organizations with distant knowledge bases to establish a common understanding and facilitate the formation of consensus on specific applications. Standardization efforts, such as developing product specifications through collective agreements, facilitate the externalization of knowledge (Nonaka, 1991) while serving as regulatory mechanisms for knowledge sharing, use, and integration in OI settings (Grøtnes, 2009).

In this respect, knowledge is not revealed but is intentionally

captured and made explicit for externalization through a pre-agreed protocol favoring its recombination. Therefore, efforts toward knowledge standardization assume strategic relevance in OI implementation as a crucial mechanism for adopting continued and intensive knowledge-based interactions in the presence of NSH attitudes (Chesbrough and Brunswicker, 2014; Zollo and Winter 2002). As an effective tool to reduce uncertainty (Shapiro and Varian, 1999), standardization efforts may counteract the negative impact of NSH attitudes by reducing information asymmetries and strengthening co-specialization among partners. From a relational perspective, efforts to codify technological specifications and procedures also epitomize the development of knowledge-sharing routines that accelerate the development of relational capital (Dyer and Singh, 1998), align partners' knowledge and expectations, and increase commitment and trustworthiness. This enhances coordination and aligns incentives and interests in knowledge sharing through the formation of consensus (Audretsch and Belitski, 2020; West and Gallagher, 2006). Thus, we contend that increasing standardization efforts will provide incentives for a higher "openness" to internal knowledge recombination by weakening the impact of negative attitudes toward knowledge sharing (NSH).

Based on these arguments, we hypothesize the following:

H4. Knowledge standardization efforts will counteract the negative impact of NSH attitudes on the depth of knowledge sourcing efforts in OI.

The conceptual model representing the relationships and interactions between the key constructs is reported in Fig. 1.

3. Data and methods

3.1. Questionnaire design and data collection

To test the proposed hypotheses, we gathered survey data on innovative Korean SMEs operating in the manufacturing industry (NACE 10–37). Because the dynamics of knowledge search and OI in services firms are substantially different from those in manufacturing industries

(Mina et al., 2014) and considering the different nature of standards in these two contexts (Paik et al., 2017), this study focused on manufacturing firms. This choice aligns with prior literature on the two topics (Blind and Mangelsdorf, 2016; Burcharth et al., 2014) and aims to add knowledge to this debate. The choice of SMEs as an empirical setting was motivated by the substantial relevance of the process of external knowledge evaluation and individual attitudes toward knowledge in this context, where decisions concerning OI depend on the positive or negative evaluation of a founder, entrepreneur, or small management team (Vanhaverbeke et al., 2018). Owing to the substantial absence of a technology portfolio, SMEs typically undertake OI as part of their larger R&D activities and with an entrepreneurial mindset (Vanhaverbeke, 2017). In this empirical context, the dynamics of external knowledge evaluation and the inherent tensions underlying search and sourcing decisions can be observed in detail compared with situations in which different subprocesses are considered within OI constructs (Du et al., 2014). As the identification of effective strategies for directly addressing knowledge tensions in OI can be challenging for organizations with limited resources, such as SMEs, leveraging standardized knowledge emerges as a promising approach to counteract NIH and NSH influence on decision-making processes. Codified knowledge facilitated by standards offers a straightforward method for evaluating and integrating external knowledge. Finally, choosing SMEs as relevant statistical units for our empirical analysis allowed us to maintain consistency with previous empirical literature investigating the impact of NIH and NSH on OI adoption and, specifically, the effectiveness of possible countermeasures (Burcharth et al., 2014).

Two different data sources, the Inno-biz list from the Korean Ministry of SMEs and Start-ups (MSS) and the R&D center database from the Korea Industrial Technology Association (KOITA), were used to assess the sampling frame. The MSS accredits innovative SMEs based on the Organization for Economic Cooperation and Development (OECD) evaluation criteria, and KOITA, on behalf of the Korean Ministry of Science and ICT (MSIT), assesses and makes a list of technology-oriented firms that have established R&D divisions. A total of 6000

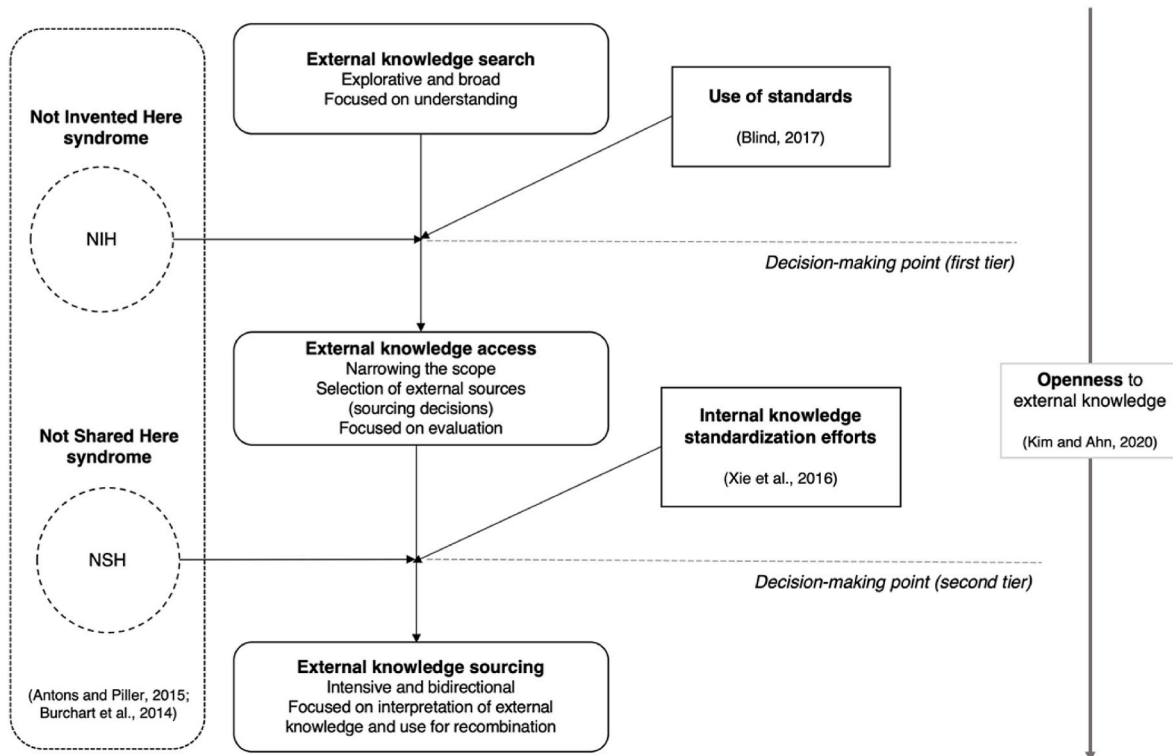


Fig. 1. Conceptual Model.

questionnaires were administered from December 2019 to January 2020 to potential respondents using an online survey platform. The questionnaire was tested by peer innovation scholars and practitioners at KOITA to refine the wording and survey structure, and it was translated from English into Korean to meet the respondents' native language. The response rate was 13.9%. Of the 834 firms, 600 met the validation criteria. In particular, 234 firms with missing values on the scale items for the main constructs were excluded from the sample, and no statistical differences in terms of industries were found between the two groups of respondents and non-respondents, indicating the absence of serious non-response biases (Armstrong and Overton, 1977).

Table 1 illustrates the sample characteristics. Firms in the sample operate in 13 different manufacturing sectors, with an average number of 71 employees, and a substantial balance among industries of different technological intensity (low-medium-high tech). The different characteristics of the industrial sectors are reflected in the share of external R&D expenditure, which is higher in low-tech sectors, and the share of R&D employees, which is higher in high-tech sectors.

We observed no substantial differences among sectoral contexts concerning the extent of OI search and sourcing efforts and the adoption of standardization practices (see Appendix, Table A1). This balance and consistency across sectors suggest that no substantial biases in the sample limit the generalizability of the results in the context of SMEs, ensuring consistency with prior research (Blind and Mangelsdorf, 2016; Burcharth et al., 2014; Paik et al., 2017).

3.2. Measurement

The scale items for the main constructs were developed using well-consolidated measures developed in the OI literature (Table A1 in Appendix A).

3.2.1. Knowledge search and knowledge sourcing in OI

Based on Katila and Ahuja (2002) two-dimensional framework of firms' innovative search, external knowledge search and sourcing decisions were defined based on two constructs, considering the scope of firms' knowledge search efforts (i.e., the degree to which firms explore new knowledge from a variety of external sources) and its intensity (i.e., the degree to which firms recombine and exploit their existing knowledge through interaction with each of these knowledge sources). The most relevant empirical extension of this framework in OI literature was followed for the operationalization of two constructs (Laursen and Salter, 2004, 2006): SCOPE considered the variety of external knowledge sources involved in firms' search processes, and DEPTH considered

the intensity of firms' knowledge-sourcing efforts from each channel.

On this basis, the two constructs were compiled into categorical variables as combinations of eight knowledge sources (Table A2). In the case of SCOPE, each item was coded as "1" when the firm reported the use of the knowledge source (and as "0" in case of no use); in the case of DEPTH, each knowledge source was coded as "1" when the firm reported a high intensity of use of that source and "0" in the case of no, low, or medium use of the given source (Laursen and Salter, 2006). SCOPE and DEPTH were then coded as the sum of the binary variables. Out of the possible range of eight external knowledge sources, the average number of sources examined by firms (SCOPE) was 2.1 (SD = 1.08), with a minimum of 0 and a maximum of 6 sources. Similarly, the average number of sources intensively used (DEPTH) was less than one (mean = 0.8, SD = 0.9) with a minimum of 0 and a maximum of 4. As the types of items included in the scale were slightly different from the original scale proposed by Laursen and Salter (2006), scale reliability was tested using the Spearman-Brown (SB) prophecy formula. Compared with Cronbach's alpha (α) coefficient, expressed as an intraclass correlation coefficient on several fixed sets of items, SB is considered a more appropriate coefficient to calculate reliability when the number of items in a questionnaire is randomly changed (De Vet et al., 2017). The value of the SB formula for the eight items considered (external knowledge sources) confirmed an acceptable value for the reliability coefficient, $r(\rho) = 0.072$.

3.2.2. NIH and NSH attitudes

Our operationalization represents an application of the theory of attitudes (Ajzen, 2001; Fabrigar et al., 2005) to the NIH framework in OI (Antons and Piller, 2015). To measure NIH and NSH, we employed IRT—a family of latent trait models advanced by research in social and cognitive psychology—for the purpose of attitudes assessment (Lord and Novick, 1968). This approach operates under the assumption that many behaviors are driven by processes operating outside of individual awareness (implicit components of attitudes) (Uhlmann et al., 2012). Capturing these implicit processes is particularly appropriate in the context of this study, as it allows for the investigation of NIH and NSH attitudes by bridging concepts and methods from psychology and cognitive science to address the main methodological limitations of current research on NIH and NSH (Antons et al., 2017).

From a methodological perspective, most existing survey measures adopt a formative approach based on attitude scales (i.e., asking direct questions about the components of attitudes, such as beliefs and intentions). However, such explicit measures of latent attitudes are subject to social desirability biases, as respondents are aware of the

Table 1
Sample characteristics.

Sectors	No. of firms	Average number of employees	R&D expenditure (share)		R&D employees (share)
			Internal R&D	External R&D	
Food/Tobacco	34	91	98.3	1.7	10.2
Textiles/Clothing	28	59	93.5	6.5	10.7
Wood/paper	18	46	95.2	4.8	12.6
Chemical	66	154	97.3	2.7	15.8
Pharmaceutical	16	83	94.7	5.3	22.9
Rubber/Plastics	28	63	95.0	5.0	9.1
Metals/Non-metallic minerals	81	51	97.4	2.6	12.6
Electronics/Communication equipment	81	61	96.0	4.0	17.7
Medical and precision equipment	19	52	96.7	3.3	21.5
Electrical equipment	49	56	96.2	3.8	18.0
Mechanical equipment	87	42	92.3	7.7	16.7
Automotive/transportation equipment	64	91	97.6	2.4	11.6
Other manufacturing sectors	29	68	93.2	6.8	22.3
Low tech	109	69	95.2	4.8	14.0
Medium tech	375	76	95.9	4.1	14.4
High tech	116	63	95.9	4.1	19.0
Total	600	72	95.8	4.2	15.2

measurement aim and may adjust their answers accordingly. As Antons et al. (2017) underline, response biases are likely to be high if questions related to attitudes are proposed in the context of NIH and NSH attitudes in OI. As external knowledge use is promoted positively in the context of OI, questions might be answered by respondents in a manner that meets expectations.

IRT-based estimates regard NIH and NSH attitudes as individual-level latent characteristics that reflect observed behavioral patterns and allow modeling the link between the target concept (latent attitude) and its observed attributes by means of reflective measures. As attitudes are reflected in the evaluative judgments of key individuals (Ajzen, 2001), this empirical examination reveals the theoretical mechanisms acting at the origin of the attitude-behavior link by assessing how observed patterns of individuals' choices toward external knowledge reflect the level of latent attitudes (NIH and NSH). Consequently, adopting the IRT framework for attitude measurement enabled us to investigate the theoretical structure of NIH and NSH and to empirically separate attitudes from their behavioral consequences. In this sense, our choice was a response to calls for research bridging concepts and methods from psychology and cognitive science in understanding NIH and NSH attitudes as the limits of OI (Dabić et al., 2023; Hannen et al., 2019).

In the framework of attitude measurement, NIH and NSH are conceptualized as attitude-based (negative) dispositions toward the acquisition and use of external knowledge, ideas, or technologies (NIH) and toward the exploitation of internal knowledge (NSH) (Katz and Allen, 1982; Lichtenthaler and Ernst, 2006; Antons and Piller, 2015; Hannen et al., 2019). To assess NIH and NSH attitudes, we used the multi-item scales designed by (Burcharth et al., 2014) to capture firm-level evaluations of external knowledge acquisition (items 1 to 3 related to NIH) and external knowledge exploitation (items 4 to 8 related to NSH) as attitude objects (Bohner and Dickel, 2011; Antons and Piller, 2015) (Table A1). As the main goal of the study was not to account for individual differences in attitudes toward external knowledge within the same firms, but rather to obtain large-scale firm-level data allowing inter-firm comparisons, we relied on the evaluative judgments of key individuals (Ajzen, 2001) and asked our respondents to perform an average firm-level evaluation of employees' attitudes toward external knowledge on a 5-point Likert scale. In the IRT framework, the two patterns of item responses to questions — firm-level evaluations of external knowledge acquisition and exploitation — mirrored the underlying latent characteristics of interest: NIH and NSH attitudes.

Because the items were measured on a 5-point scale, a graded response model (GRM) denoted as (IRT-GRM) was employed (Marullo et al., 2022). This model specifies the probability that firm i will provide a response corresponding to the k th category of each item j related to external knowledge evaluation (π_{ijk}). This probability is conditional on the level of an unobservable attitude (θ_i) (Samejima, 1969):

$$P(Y_{ij} \geq k | \theta_i) = \frac{\exp(\alpha_j \theta_i - k_{j,k})}{1 + \exp(\alpha_j \theta_i - k_{j,k})}$$

In the proposed framework, the subjective evaluation of external knowledge is conditional on the levels of NIH (θ_{NIH}) and NSH (θ_{NSH}) (i.e., the firm's underlying attitude). The unidimensional latent measures resulting from the model are assessed by two continuous variables, noted as θ_{NIH} and θ_{NSH} , with a mean of 0 and a standard deviation (SD) of 1, and an arbitrary range covering the latent trait that is measured (in this case, θ_{NIH} and θ_{NSH} both range from -4 to 4 , with firms closer to -4 having lower NIH/NSH attitudes and firms closer to 4 having higher NIH/NSH attitudes). The test information function curves and standard errors reported in the Appendix (Figure B1 and B2) show that the three items in the hypothetical NIH scale and the four items in the hypothetical NSH scale can discriminate firms over a wide range of latent measures θ_{NIH} and θ_{NSH} ($+3/-3$). Thus, the test confirms that the two constructs based on the scales developed by (Burcharth et al., 2014)

have good measurement validity (Yang and Kao, 2014). For ease of interpretation, the two measures were rescaled from 0 to 100 (θ_{NIH} mean = 46.3, SD = 15.1; θ_{NSH} mean = 56.3, SD = 17.4).

3.2.3. Use of standards and standardization efforts

To investigate the role of standardization as an indirect counter-measure to NIH and NSH attitudes emerging in the OI implementation process, two variables were operationalized. Based on prior empirical studies by Blind and Mangelsdorf (2016) and Paik et al. (2017) on standardization in Korean SMEs, STD_use was operationalized as a binary variable (1 = yes, 0 = no) related to the use of almost one of the standards developed by the following standards-setting organizations (SSOs): International public standards (ISO, IEC, ITU), international private standards (IEEE, ASTM, ASME, UL, DIN, BS, CSA, JEC), domestic national standards (Korean standards), domestic SDOs (standard development organizations) standards, and company (in-house) standards. Standardization efforts (STD_effort) were measured based on the respondents' evaluation of the top management or R&D department head's actual commitment toward the design and development of standards over internal knowledge on a 5-point Likert scale (mean = 3.7, SD = 0.8).

3.2.4. Control variables

Control variables related to firms' structural characteristics and prior collaborative innovation activities were considered. Firm size, measured by the total number of employees (Employees_num), showed high overdispersion (mean = 53.6, SD = 56.5), as 65% of the sample contained SMEs with fewer than 50 employees and only 4% had more than 200 employees. External R&D investments (EXT_R&D), a proxy for firms' efforts toward collaborative innovation activities (e.g., research and development, cooperation, and IP in-licensing), were measured as the ratio of external to total R&D expenditure in the last three years (mean = 4.2; SD = 12.5). Finally, 13 industry dummies were considered to control for interindustry differences in internal R&D intensity and their influence on the degree of openness in firms' search strategies (Huizingh, 2011; Laursen and Salter, 2006).

4. Empirical analysis and results

4.1. Estimation and model specification

To test the research hypotheses, we employed a hurdle model for count data using a two-step estimation procedure. The hurdle model loosens the assumption that the zeros and positive values in a distribution come from the same data generation process; as such, it considers a two-step decision-making process, in which each part is a model of one sub-process (Cameron and Trivedi, 1986). In our case, because the outcome of interest (DEPTH) was a discrete, non-negative integer (count) variable with many zeros, the hurdle model integrated a standard binary outcome model (binary logit) in the first tier, to estimate the probability of a zero-versus-positive outcome - i.e. the probability of transition from broad search to external knowledge sourcing [$\Pr(\text{DEPTH} = 1)$]. In the first tier, D_DEPTH^* denotes a latent variable accounting for the fact that the outcome (DEPTH) is observed only if $D_DEPTH^* > 0$. A zero-truncated Poisson function was used in the second tier based only on the sample with positive observations to model the value of the dependent variable conditional on a positive count (DEPTH). Thus, the first tier of the hurdle model considers firms' decisions to transition from broad explorative search (SCOPE) toward engagement in external knowledge sourcing (D_DEPTH^*), while the second tier considers firms' decisions to increase the intensity of external knowledge sourcing efforts (DEPTH). A common set of regressors was used in both parts of the model, and the parameters and correlation terms were obtained using Maximum Likelihood (ML) estimation techniques.

To limit concerns about confounding effects due to common method

variance, we ran Harman's single-factor test as a post hoc analysis (Podsakoff et al., 2003). An exploratory factor analysis was conducted for all variables and controls considered in the study, and the unrotated factor solution was examined to determine the number of factors accounting for most of the covariance in the measures. As the variables were not loaded into a single general factor (seven factors emerged from the factor analysis) and no general factor accounted for most of the covariance among the measures (the factor showing the highest variance explanation accounted for 11% of the variance in the dataset), we concluded that common method variance was not a serious issue. Table 2 presents the descriptive statistics and correlation matrices for the variables and controls. Considering the observed correlations, we ran separate models with unstandardized and standardized coefficients to avoid multicollinearity (Cronbach, 1987; Dunlap and Kemery, 1987) and to interpret first-order effects in the presence of interactions (Aguinis and Gottfredson, 2010). Because of standardization, the Variance Inflation Factor (VIF) reported a significant reduction, as VIF values for all variables fell below 2 (mean VIF = 1.02). The standardized coefficients also allowed us to compare the magnitudes of the estimated coefficients by removing the influence of different measurement units.

4.2. Empirical results

Table 3 reports the hurdle model estimates. The relationship between the latent level of NIH attitudes (θ_{NIH}) and D_DEPTH* in the first tier was modeled through binary logit functions, and the influence of STD_use in counteracting the expected negative impacts of NIH on D_DEPTH* was tested by examining the main and moderation effects. In the second-tier regressions, a zero-truncated Poisson function was used to test the hypotheses related to the intensity of knowledge sourcing efforts (i.e., based only on the sub-sample of firms with positive values of DEPTH). The expected negative influence of the latent level of NSH attitudes (θ_{NSH}) on DEPTH and the influence of STD_effort on counteracting NIH were tested through main effects and moderation effects. Because our procedure included a test of four moderating effects, a hierarchical regression approach was adopted by introducing controls, independent variables, moderators, and interaction terms in separate steps (Models 1–8).

First-tier regressions of the hurdle model estimate D_DEPTH* (Models 1–4). In all the models, the estimated coefficients for SCOPE and EXT_R&D are positive and significant, indicating that systematic efforts toward broadening the scope of external knowledge search increase the probability of engagement in more intensive knowledge sourcing practices and are reinforced by external R&D investments (Cassiman and Veugelers, 2006; Laursen and Salter, 2006). More in detail, the main effects models (Models 1–2) show a negative and significant coefficient for the estimated level of latent NIH (θ_{NIH}), confirming the impact of NIH attitudes in limiting external knowledge sourcing decisions (Burcharth et al., 2014) and a positive and significant coefficient for firms using standards ($\beta_{STD_use} = 0.603$, $p = 0.006$), whereas no significant effect is observed for STD_efforts at this stage.

Models 3 and 4 test the moderating effects of STD_use and STD_effort on the relationship between θ_{NSH} and D_DEPTH*. The results reveal that

the use of standardized knowledge inputs (STD_use) enables firms to move forward from a broad explorative search to external knowledge sourcing, thus countering the negative impact of θ_{NIH} on external knowledge evaluation and sourcing decisions ($\beta_{\theta_{NIH} \# STD_use} = 0.325$; $p = 0.10$, Model 3). As firms confront an increasing variety of external knowledge sources, the use of standards facilitates decisions based on the evaluation of external knowledge by attenuating the cognitive distance between internal and external knowledge domains, favoring the positive evaluation of internal knowledge over external sources (West et al., 2014). As the main effects are also significant ($\beta_{STD_use} = 0.645$ $p = 0.004$); STD_use can also be considered as a partial moderator (i.e. the probability of transition significantly increases for firms using standards from national or international SSOs in presence of NIH attitudes).

Models 5–8 report the results of the second-tier regression of the hurdle model, estimating the value of the outcome variable (DEPTH) conditional on the subsample with positive count observations ($D_DEPTH^* > 0$, $n = 305$). The results for the main effects models (Models 5–6) show a negative and significant coefficient for θ_{NSH} , indicating that NSH attitudes negatively impact decision-making on the intensity of external knowledge sourcing efforts (DEPTH). Hence, while NIH attitudes represent prominent barriers to external knowledge sourcing decisions, the impact on NSH attitudes on the depth of sourcing efforts once the transition occurs is a topic of strategic relevance in OI implementation, as it determines “how frequently the firm reuses its existing knowledge” (Katila and Ahuja, 2002, p. 1183). In the second tier, standardization efforts (STD_effort) are positively and significantly associated with the intensity of external knowledge sourcing (DEPTH). At this stage of OI implementation, setting the incentives for the dynamic process of knowledge recombination, STD_use does not show any significant effect. The degree of firms' proactive effort toward knowledge standardization (STD_effort) plays a significant role in enhancing the depth of external knowledge sourcing from external partners. Finally, model 8 confirms the significant moderating effect of STD_effort on the relationship between θ_{NSH} and DEPTH ($\beta_{\theta_{NSH} \# STD_efforts} = 0.153$ $p = 0.05$). This evidence confirms that, in the presence of NSH attitudes, efforts toward proactive knowledge standardization will enable a shared understanding of types of knowledge that will be most relevant. Through the formation of consensus, standardization reduces uncertainty and strengthens co-specialization (Teece, 1986), thus counteracting the behavioral impact of NSH attitudes on OI implementation.

5. Discussion

Viewing the implementation of OI as a dynamic process in which the evaluation of increasingly complex and heterogeneous knowledge is a core managerial challenge, this study examined the role of standardization in debiasing external knowledge search and sourcing decisions in the presence of NIH and NSH attitudes.

First, we examined the mechanisms through which NIH and NSH, as individual-level sources of knowledge-related tensions, have negative behavioral consequences in the sub-processes of searching, accessing, and using external knowledge for recombination, which are at the core of the OI paradigm (Chesbrough 2019). An important conjecture of this

Table 2
Descriptive statistics and correlations.

	N	mean	sd	min	max	Correlation matrix							N. employees
						SCOPE	DEPTH	θ_{NIH}	θ_{NSH}	STD_use	STD intensity	External R&D share	
SCOPE	600	0.8	0.9	0	4	1.00							
DEPTH	600	2.1	1.1	0	6	0.36	1.00						
θ_{NIH}	600	46.3	15.1	0	100	−0.13	−0.04	1.00					
θ_{NSH}	600	56.3	17.4	0	100	−0.09	0.01	0.32	1.00				
STD_use	600	0.7	0.5	0	1	0.13	0.16	0.04	0.09	1.00			
STD_effort	600	3.7	0.9	2	5	0.17	0.16	−0.04	−0.06	0.33	1.00		
EXT_R&D	600	4.2	12.5	0	83.7	0.16	0.11	−0.02	−0.12	0.04	0.19	1.00	
Employees_num	600	53.6	56.5	1	249	0.07	0.15	0.00	0.06	0.16	0.13	0.05	1.00

Table 3
Hurdle model estimates.

Dependent variables	First-tier regressions: (binary logit models)				Second-tier regressions (zero-truncated Poisson models)			
	1	2	3	4	5	6	7	8
	D_DEPTH*	D_DEPTH*	D_DEPTH*	D_DEPTH*	DEPTH	DEPTH	DEPTH	DEPTH
SCOPE	0.497 ^a	0.454 ^a	0.460 ^a	0.456 ^a	0.451 ^a	0.478 ^a	0.470 ^a	0.482 ^a
θ_{NIH}		−0.326 ^a	−0.557 ^a	−0.698 ^a		−0.032	−0.032	−0.012
θ_{NSH}		0.019	0.015	0.027		−0.190 ^b	−0.389 ^b	−0.278 ^a
STD_use		0.603 ^a	0.645 ^a	0.596 ^a		−0.071	0.071	0.023
STD_effort		0.074	0.069	0.080		0.159 ^c	0.179 ^b	0.192 ^b
$\theta_{NIH} \# \text{STD_use}$			0.325 ^c					
$\theta_{NIH} \# \text{STD_effort}$				0.099				
$\theta_{NSH} \# \text{STD_use}$							0.253	
$\theta_{NSH} \# \text{STD_effort}$								0.153 ^b
EXT_R&D	0.283 ^a	0.265 ^a	0.267 ^a	0.263 ^b	0.067	0.024	0.019	0.026
Employees_number	0.062	0.014	0.206	0.018	−0.004	−0.011	−0.021	−0.103
INDUSTRY DUMMIES	YES	YES	YES	YES	YES	YES	YES	YES
<i>Model statistics</i>								
N	600	600	600	600	305	305	305	305
Wald Chi-2	64.27	86.24	88.76	87.12	122.79	134.02	135.58	137.91
Model fit	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Cox and Snell R ²	0.101	0.134	0.137	0.135	0.218	0.243	0.247	0.252
Pseudo R ²	0.077	0.104	0.107	0.105	0.196	0.214	0.217	0.221

^a Significant at 0.01.

^b Significant at 0.05.

^c Significant at 0.1.

study is that increasing openness to external knowledge is not the result of distinct strategic choices (Barge-Gil, 2010; Huizingh, 2011) but rather a process of learning, requiring a time-consistent positive evaluation of external knowledge to facilitate decision-making (Kim and Ahn, 2020; Marullo et al., 2021). From this perspective, it is surmised that decision making at the organizational level may be biased in the presence of NIH and NSH attitudes at different stages (subprocesses), and that this requires appropriate countermeasures at each stage. This study particularly focused on clarifying the mechanisms through which NIH and NSH attitudes emerge with increasing openness to external knowledge (Audretsch and Belitski, 2023; Dabić et al., 2023). We showed that higher levels of NIH and NSH have negative behavioral consequences on OI implementation because they represent individual-level sources of tension with cascading effects on access to external knowledge and the intensity of external knowledge sourcing. While NIH attitudes limit external knowledge access and evaluation leading to a bounded search, NSH attitudes are more influential when knowledge sourcing increases in intensity, leading to inward decisions (Antons and Piller, 2015; Purdy et al., 2023; West et al., 2014). Our empirical evidence confirms that the “bright side” of OI (i.e., when the benefits of broad explorative search strategies are realized in external knowledge sourcing decisions, and then lead to more intensive sourcing activities) may be offset by the “dark side” of OI as NIH and NSH attitudes emerge, causing errors in knowledge evaluation and influencing the incentives for knowledge recombination, leading to reluctance in moving forward in this process. We provide evidence of these phenomena in the context of SMEs, which typically reap greater benefits from opening up to external knowledge flows compared with larger firms (Brunswick and Vanhaverbeke, 2015) but face severe challenges in external knowledge evaluation, adoption, and integration (van de Vrande et al., 2009; Vanhaverbeke et al., 2018). We observed the emergence of knowledge challenges in different phases of the OI process. In the first tier of the hurdle model, the transition from broad explorative search to knowledge sourcing activities is observed only for a subsample of SMEs and the coefficients of the latent variable related to NIH (θ_{NIH}) are significant. This suggests that negative dispositions toward external knowledge negatively impact

sourcing decisions, fostering the rejection of potentially relevant knowledge for value creation (i.e., limiting firms’ interest to move forward in OI implementation). Interestingly, the estimated coefficients of NSH attitudes (θ_{NSH}) become significant in the second tier of the hurdle model, which predicts the intensity of external knowledge sourcing. This confirms that knowledge tensions represent critical hurdles as SMEs decide to engage in deeper sourcing (Brunswick and van de Vrande, 2014; van de Vrande et al., 2009; West, 2006). These findings also suggest that NIH and NSH attitudes may lead to biased decision making at different stages of OI implementation (requiring increasing openness) and become critical as sourcing efforts need to be balanced with selective revealing and asset control (Marullo et al., 2020, 2021; Oakey, 2013).

Second, this study examined the role of knowledge standardization in counteracting NIH and NSH attitudes (i.e., its role as a debiasing mechanism) in this process. Our findings suggest that the use of standardized knowledge inputs and the effort put into the design and collaborative implementation of knowledge standardization practices are significant in enabling the transition process. While firms using standards in the process of external knowledge search experience a lower negative impact of NIH on sourcing decisions, only those with persistent standardization efforts eventually reduce the impact of NSH when increasing the intensity of knowledge sourcing. This different use of standardization practices— reactive or proactive— aligns with the nature of knowledge search and sourcing decisions in OI. While the decision to shift from broad search to knowledge sourcing (i.e., a progressive focus on OI implementation with increased strategic relevance) implies a process of evaluating heterogeneous knowledge from different sources, the use of standardized knowledge inputs is effective in counteracting the influence of NIH attitudes (Antons and Piller, 2015). However, once deep and focused external knowledge sourcing becomes the dominant strategy, requiring a commitment to knowledge exchange with external partners for recombination, concerns about possible knowledge leakage would act as a hurdle (Ritala et al., 2015). In this context, the proactive use of standardization (i.e., increasing standardization efforts) would not only decrease uncertainty and lower

information asymmetries but also strengthen the intensity of links with network partners (Teece, 1986). By attenuating the attitude-behavior relationship, the pursuit of standardization—a regulatory mechanism for knowledge sharing, use, and diffusion (Xie et al., 2016)—in OI settings promises to be effective in counteracting the emerging “limits” of OI, specifically the negative effects of NIH and NSH attitudes on external knowledge evaluation and use for recombination (Audretsch and Belitski, 2023; Hannen et al., 2019).

6. Conclusions

The implementation of OI involves complex decisions based on the evaluation of external knowledge, and requires the development of organizational culture and management systems that allow the face knowledge-related challenges at multiple levels (Chiaroni et al., 2010; Kim and Ahn, 2020). With some relevant exceptions (e.g., Audretsch and Belitski, 2020; Chiaroni et al., 2011; Stefan et al., 2022), the literature has explored the “limits to OI” predominantly from a static perspective, and focusing mostly on single levels of analysis (i.e., individuals or organizations). By adopting a multi-level perspective, this study observed the mechanisms through which NIH and NSH attitudes as individual-level sources of knowledge tensions in OI may produce cascading effects on search and sourcing decisions, (Dabić et al., 2023; Ritala and Stefan, 2021; Stefan et al., 2022). Furthermore, it provides empirical evidence of the effectiveness of standardization as a possible indirect countermeasure to knowledge tensions in OI, intended to contain the negative behavioral effects of NIH and NSH rather than aiming at reducing or changing these attitudes (Hannen et al., 2019).

In addressing the complex managerial challenges involved in evaluating external knowledge within OI processes, this study provides further insight into how misaligned incentives due to NIH and NSH emerge, are identified, and addressed by firms (West et al., 2014), thus extending the current understanding of the micro foundations of OI challenges (Hannen et al., 2019; Ritala and Stefan, 2021).

Responding to recent calls for research that bridges concepts and methods from psychology and cognitive science in understanding the challenges of OI (Bogers et al., 2017; Dabić et al., 2023), we explored the influence of NIH and NSH attitudes on decision-making at different stages of OI adoption and empirically examined how and when the pursuit of knowledge standardization is effective in “de-biasing” external knowledge search and sourcing decisions. Therefore, our study expands the literature on OI by proposing and testing the role of standardization in counteracting the behavioral effects of NIH and NSH attitudes, thereby contributing to the debate on solutions to the managerial challenges of accessing, evaluating, and using external knowledge in OI. This study not only provides a portrayal of the dynamics of OI and its challenges but also offers a new perspective at the intersection between knowledge standardization and OI literature.

The study also demonstrated that the pursuit of standardization acts as an enabler in the evolution of firms’ external knowledge search and sourcing strategies. By facilitating access to and evaluation of potentially relevant external knowledge, aiding its interpretation and application in different contexts, and supporting recombination through the formation of collective consensus, this study shows that standardization efforts help debias knowledge search and sourcing decisions in the presence of NIH and NSH. Thus, it adds to the body of literature on the relevance of firms’ standardization efforts for knowledge search and innovation (Xie et al., 2016) by offering new insights into the role of standards in facilitating OI processes.

From a managerial perspective, CEOs, CTOs, and entrepreneurs should be aware that the adoption of OI is not the result of a binary strategic choice (i.e., “do, or do not”), but a stepwise process, demanding different types of managerial support to counteract knowledge challenges at different stages. This study highlights the importance of knowledge standardization in mitigating possible decision-making errors caused by tensions when confronted with external knowledge. It

also suggests that an optimized fit should be established to address different types of tensions at different stages of the process. In line with prior research employing a process view of OI implementation (Kim and Ahn, 2020) this study suggests that higher managerial efforts are necessary as the degree of openness toward external knowledge increases. While “trial and error” explorative approaches in the initializing stage of OI projects may be pursued without significant difficulties, knowledge integration demands substantial organizational efforts (Marullo et al., 2020). This view provides a clear description of how OI implementation challenges occur in practice and of the possible countermeasures, as increasing openness to external knowledge requires a decision-making process based on a consistent evaluation of external knowledge at each stage.

Another relevant implication for managers and entrepreneurs, particularly those operating in SMEs, is that greater attention should be paid to the efficacy of pursuing standardization to facilitate open and collaborative innovation processes. In situations where there is a lack of internal competencies to evaluate external knowledge, which represent typical managerial challenges to OI in SMEs (van de Vrande et al., 2009), search and sourcing strategies might be biased if NIH and NSH are used as information processing heuristics (Antons and Piller, 2015). Unlike their larger counterparts, SMEs may benefit more from indirect countermeasures based on knowledge standardization rather than adopting costly structural solutions aimed at reducing NIH and NSH. In the context of SMEs, standards might prove effective in attenuating the negative behavioral consequences of such attitudes. This study demonstrates that the use of standards in external knowledge search can simplify access and interpretation of relevant external knowledge, reduce search and evaluation costs, and potentially enhance external knowledge absorption capabilities. Furthermore, for internal knowledge exploitation, the study suggests that a proactive use of knowledge standardization practices in OI settings facilitates selective revealing and promotes co-specialization with external partners.

Given the significant economic function of standards, our study also offers meaningful policy implications. Traditionally, standardization has been viewed primarily as a regulatory tool aimed at enhancing the diffusion of technologies (Blind, 2017). However, this study suggests that promoting knowledge standardization efforts might represent a relevant policy tool for fostering OI, particularly in SMEs (Bogers et al., 2018). Globally, governments have attempted to promote co-design and collaboration in innovation between organizations, mainly through R&D incentives. Although direct R&D funding is an effective intervention for strengthening a firm’s internal knowledge base, its positive impact could decrease if excessive resources are allocated to firms (Ahn et al., 2020). Given these ambiguities, policymakers should consider promoting standardization practices as a complementary policy tool to more traditional R&D incentives, addressing the complexities of knowledge evaluation and use in OI partnerships.

Given its exploratory nature, this study had several limitations that represent opportunities for further research. First, we used cross-sectional data for the empirical analysis. Although we used three-year averaged survey data to account for time lags in the stepwise process, future research should consider using longitudinal datasets to further investigate the causal relationships between key variables over time and to account for any changes in direction. Future research could also explore the influence of standardization practices on new product development and innovation performance once the main challenges are overcome. Second, this study deliberately focused on manufacturing SMEs. Given the distinct dynamics of external knowledge search for OI in the services industry and the unique nature of standards within these contexts, our findings may not directly translate to services. Future research on knowledge challenges in OI would require a different theoretical framework and empirical approach for service firms. Third, as this study focuses on Korean firms, some results may reflect unspecified contextual elements that vary across countries. Future research should examine whether the findings of this study can be applied to

other countries with different national innovation systems.

Data availability

The authors do not have permission to share data.

CRediT authorship contribution statement

Cristina Marullo: Conceptualization, Methodology, Formal analysis, Writing – original draft, Writing – review & editing. **Joon Mo Ahn:** Conceptualization, Data curation, Project administration, Resources, Writing – original draft, Writing – review & editing.

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Appendix A

Table A1

Descriptive statistics on OI adoption and standardization among firms in the sample

Sectors	No. of firms	Scope of external search	Depth of sourcing efforts	Use of Standards (share of firms)	Standardization effort (average, 1 to 5)
Food/Tobacco	34	2.1	1.4	3.9	3.5
Textiles/Clothing	28	2.4	1.6	2.8	3.4
Wood/paper	18	1.9	1.5	3.5	3.6
Chemical	66	2.0	1.7	10.9	3.4
Pharmaceutical	16	2.1	1.6	3.2	4.2
Rubber/Plastics	28	1.8	1.4	5.3	3.9
Metals/Non-metallic minerals	81	1.8	1.3	14.6	3.7
Electronics/Communication equipment	81	2.3	1.7	13.3	3.8
Medical and precision equipment	19	2.4	2.0	3.9	3.9
Electrical equipment	49	2.0	1.4	9.1	3.9
Mechanical equipment	87	2.0	1.4	15.6	3.9
Automotive/transportation equipment	64	2.1	1.3	11.6	3.6
Other manufacturing sectors	29	2.0	1.3	2.1	3.4
Low tech	109	2.1	1.5	12.3	3.4
Medium tech	375	2.0	1.4	62.5	3.7
High tech	116	2.3	1.7	19.3	3.9
Total	600	2.1	1.5	100	3.7

Table A2

Scale items for the main constructs

Construct	Source	Questions and items	Measure
SCOPE	Laursen and Salter, 2004, 2006	Where did your company obtain relevant information and knowledge for innovation in the past year?	Importance (0 = no use; 1 = low; 2 = medium; 3 = high)
DEPTH		1) Universities or other higher education institutions 2) Competitors/other firms from the same industry 3) Government/private non-profit research institutes 4) Commercial laboratories/R&D enterprises/Consultants 5) Suppliers of equipment, materials, components or software 6) Clients or customers 7) Professional/industry associations, other organizations 8) Technical/trade press, databases, fairs, exhibition	
Not Invented Here (NIH) attitude	Burcharth et al. (2014)	Please evaluate the following statements regarding your company's knowledge: Our employees are negative about the application of outside ideas and technologies to our products and services. Our employees believe that external ideas and technologies are just as valuable as internal ones*.	Relevance (1–5)
Not Shared Here (NSH) attitude	Burcharth et al. (2014)	Our employees have often received knowledge from outside and used it*. Our employees are negative about the acceptance and use of our knowledge and skills by other companies Our employees are not reluctant to commercialize our technology by external partner companies*. Our employees have often sold or disclosed our knowledge and skills to other companies*. Our employees are positive about developing new ideas, solutions and technologies for other companies*.	Relevance (1–5)
Use of standards	Blind and Mangelsdorf, 2016; Paik et al., 2017	Have you used (viewed or purchased) the standards of the following SSOs (Standard Setting Organizations)?	Use (yes, no)

(continued on next page)

Table A2 (continued)

Construct	Source	Questions and items	Measure
Standardization effort	Blind and Mangelsdorf, 2016; Paik et al. (2017)	International public standards (ISO, IEC, ITU etc)	Importance (1–5)
		International private standards (IEEE, ASTM, ASME, UL, DIN, BS, CSA, JEC etc)	
		Domestic national standards (Korean Standard)	
		Domestic SDOs (Standard Development Organizations) standards	
		Please evaluate your (the top management or R&D department head's) actual commitment towards knowledge standardization activities	

*reverse coded.

Appendix B

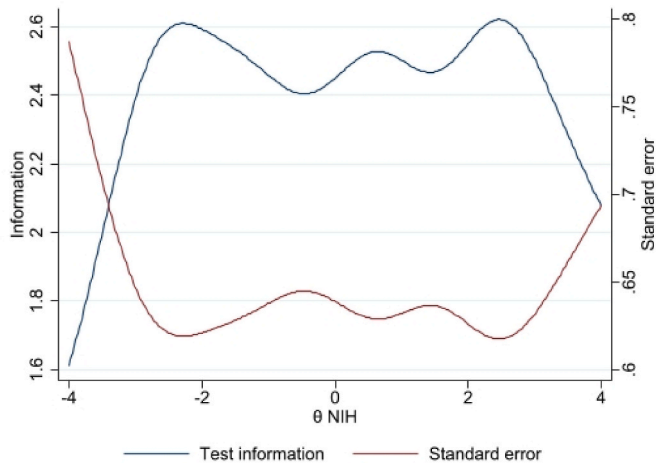


Fig. B1. Test information function (TIF) for θ_{NIH}

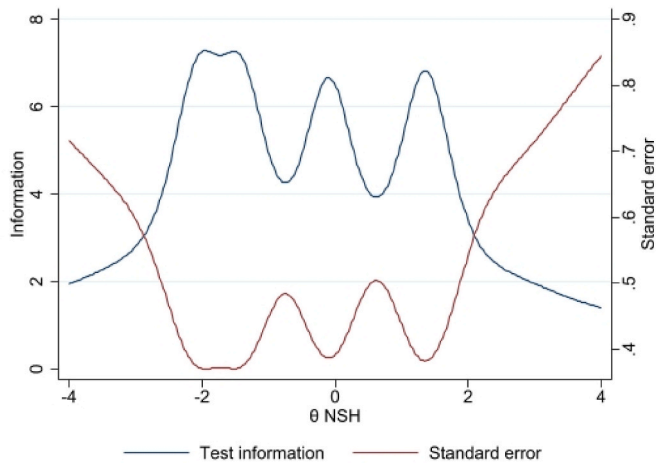


Fig. B2. Test information function (TIF) for θ_{NSH}

Appendix C

Robustness analyses

Additional robustness analyses were conducted to confirm the results. A first robustness check was performed by issuing a specification link test for single-equation models. to test the adequacy of the chosen link function (logit) in the two tiers of the hurdle model (first tier: logistic regression; second tier: zero-truncated Poisson regression). The test for model specification also allowed us to test the relevance of the predictors and their linear combination in the two tiers of the model. As the statistical models were fit using the maximum likelihood method, we applied likelihood chi-square tests and calculated the Akaike information criterion at each step of estimation. Allowing us to assess their goodness of fit. We further issued log-likelihood chi-square tests and calculated the Akaike information criterion at each step of estimation to assess the goodness of fit of the model. Wald chi2 and joint significance tests were run to assess the strength of the interaction effects and the significance of the coefficients. Finally. Robustness checks were performed by examining alternative estimation techniques.

Specification error

A first diagnostic analysis was conducted by issuing a specification link test for single-equation models. to a) test the model adequacy. Assuming that the logit link is the correct function to use. And b) test the relationship between the logit of the outcome variable and the independent variables. to exclude the hypothesis that their combination is not linear.

The results of the specification link test for model adequacy are reported in [Table C1](#).

The linear predicted value from the model -testing its adequacy-is statistically significant in both the first and the second-tier regressions (binary logit and zero-truncated Poisson models).

This indicates that the logit link function is correctly specified. On the other hand. As the square of the linear predicted value (link test) is not significant. the null hypothesis (we have omitted relevant variables) is rejected. This leads to conclude that the model includes meaningful predictors (and their interactions) and their combination to the link function is linear.

Table C1

– Results of the specification link tests

	First tier regressions (binary logit model)	Second tier regressions (zero-truncated Poisson model)
Linear predicted value	1.040***	1.051***
Squared linear predicted value	−0.198	−0.179
constant	0.106	0.0692

***p < 0.001; **p > 0.01; *p < 0.05.

Goodness of fit

As the statistical models were fit using the maximum likelihood method. we applied likelihood chi-square tests and calculated the Akaike information criterion at each step of estimation. Allowing us to assess their goodness of fit.

Based on the ratio of the log likelihoods at each step. Likelihood chi-square tests allowed us to compare the goodness of fit of alternative - restricted and unrestricted-model specifications i.e. those including the controls. the independent variables. the moderator variables. And the interaction terms in each tier. The likelihood ratio test allowed to check if the restricted model specification implied a statistical model that is nested within the model implied by the subsequent -unrestricted - specifications.

Akaike's information criterion (AIC) allows for comparing models based on the log-likelihood estimated at each step.

The criterion is defined as:

$$AIC = -2 \ln L + 2k$$

where $\ln L$ is the maximized log-likelihood of the model and k is the number of parameters estimated. Differently from the likelihood square ratio test models do not need to be nested to be compared.

A higher (and significant) value of the likelihood ratio chi2 test. And a lower value of the AIC indicate an improvement in the goodness of fit on subsequent model specifications. The results of tests applied to the two tiers of the hurdle model show that adding the independent variables together (Model 2 and Model 6) and the interaction terms (Models 3–4 and Models 7–8) in each tier significantly improves the fit of the model. compared to the restricted model specifications (Model 1 in the first tier and Model 5 in the second tier).

This result also confirms that the models are fitted with the same estimation command (i.e., by a binary logit in the first tier and by a zero-truncated Poisson regression in the second tier. Both using the logit link function). And that the subset of covariates of one model is a subset of the covariates of the subsequent model.

Table C2

– Results of the likelihood ratio tests and AIC values

	LR chi2	AIC
Model 1	–	799.34
Model 2	21.97***	785.37
Model 3	24.48***	784.85
Model 4	22.84***	786.49
Model 1	–	534.76
Model 6	11.23*	531.53
Model 7	12.79*	531.98
Model 8	15.12**	529.64

***p < 0.001; **p > 0.01; *p < 0.05.

Finally. we tested the significance of the effects of the two latent attitude measures (θ_{NIH} . θ_{NSH}). And of the variables assessing standardization practices (STD_use and STD_efforts). considering interactions. through a) a comparison of Wald chi-square statistics without vs. with interaction and b) joint significance Wald tests.

The strength of the interaction effect is reflected in the difference between the Wald chi2 statistic of models without vs. with interaction.

Further. As the joint significance test of the effects of θ_{NIH} and STD_use and θ_{NSH} and STD_efforts. taking into account interactions. Reports reported significant p values (p < 0.01). we concluded that the null hypothesis (the coefficients are equal to zero) had to be rejected.

Table C3
– Results of the Wald tests

	Wald chi2	Joint significance Wald test
θ_{NIH} - STD_use	86.24***	17.96***
θ_{NIH} # STD_use	88.76***	15.50***
θ_{NSH} - STD_effort	134.02***	9.96**
θ_{NSH} # STD_effort	137.91***	12.13**

***p < 0.001; **p < 0.01; *p < 0.05.

Alternative estimations

Further robustness checks were performed on the estimations by examining alternative techniques. Models 9–16 provide estimates based on binary logit models and zero-truncated Poisson models. using robust standard errors. Models 17–20 provide estimates based on zero-truncated negative binomial regression models (using robust standard errors). The results of the models (including interactions) obtained by using different estimation techniques were qualitatively consistent with those of the corresponding models obtained by using the original models (Table C4. Models 1–8).

Table C4
– Robustness analyses

Dependent variables	First-tier regressions (binary logit models)				Second-tier regressions (zero-truncated Poisson models)							
	9	10	11	12	13	14	15	16	17	18	19	20
	D_DEPTH*	D_DEPTH*	D_DEPTH*	D_DEPTH*	DEPTH	DEPTH	DEPTH	DEPTH	DEPTH	DEPTH	DEPTH	DEPTH
SCOPE	0.497***	0.454***	0.460***	0.456***	0.451***	0.479***	0.471***	0.482***	0.419***	0.479***	0.471***	0.482***
θ_{NIH}		–0.326***	–0.557***	–0.698***		–0.032	–0.032	–0.012		–0.032	–0.032	–0.012
θ_{NSH}		0.019	0.015	0.027		–0.190***	–0.389**	–0.278***		–0.190**	–0.389**	–0.278***
STD_use		0.603***	0.645***	0.596***		–0.071	0.071	0.023		–0.706	0.071	0.050
STD_efforts		0.074	0.069	0.080		0.159*	0.179**	0.192**		0.159**	0.179**	0.192**
θ_{NIH} x STD_use			0.325*									
θ_{NIH} x STD_efforts				0.099								
θ_{NSH} x STD_use							0.253*				0.253*	
θ_{NSH} x STD_efforts								0.153**				0.153**
EXT_R&D	0.283***	0.265***	0.267***	0.263***	0.067	0.024	0.019	0.026	0.005*	0.024	0.019	0.026
Employees_num	0.062	0.014	0.206	0.018	–0.004	–0.011	–0.021	–0.103	0.001	–0.011	0.021	0.010
INDUSTRY DUMMIES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Model statistics												
N	600	600	600	600	305	305	305	305	305	305	305	305
Chi-square	60.88	76.98	79.75	77.62	177.30	177.30	177.80	177.90	120.46	177.30	167.68	167.98
Model fit	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Cox & Snell R2	0.101	0.134	0.137	0.135	0.214	0.243	0.247	0.252	–	–	–	–
Pseudo R2	0.077	0.104	0.107	0.105	0.196	0.214	0.217	0.221	0.102	0.123	0.217	0.220

*** Significant at 0.01; ** Significant at 0.05; * Significant at 0.1 Models specification: LOGIT. robust (models 9–12); ZTP. robust (models 13–16); ZTNB. robust (models 17–20).

Appendix D

Table D1
Marginal effects of the interaction between θ_{NSH} and STD_use (Model 3)

Pr(INTENSITY) = 1	at#STD_use	Margin	Std.Err.	z	P > z	[95%Conf.	Interval]
at $\theta_{NIH} = -2.0$	no	0.693	0.091	7.650	0.000	0.515	0.870
	yes	0.683	0.062	11.000	0.000	0.561	0.805
at $\theta_{NIH} = -1.5$	no	0.619	0.077	8.050	0.000	0.468	0.770
	yes	0.652	0.051	12.870	0.000	0.553	0.752
at $\theta_{NIH} = -1.0$	no	0.540	0.059	9.160	0.000	0.424	0.655
	yes	0.621	0.039	16.000	0.000	0.545	0.697
at $\theta_{NIH} = -0.5$	no	0.459	0.042	10.830	0.000	0.376	0.542
	yes	0.588	0.028	20.950	0.000	0.533	0.643
at $\theta_{NIH} = 0$	no	0.380	0.036	10.450	0.000	0.309	0.452
	yes	0.555	0.023	24.220	0.000	0.510	0.600

(continued on next page)

Table D1 (continued)

Pr(INTENSITY) = 1	at#STD_use	Margin	Std.Err.	z	P > z	[95%Conf.	Interval]
at $\theta_{NIH} = 0.5$	no	0.308	0.043	7.200	0.000	0.224	0.391
	yes	0.521	0.028	18.670	0.000	0.467	0.576
at $\theta_{NIH} = 1$	no	0.243	0.052	4.700	0.000	0.142	0.344
	yes	0.487	0.040	12.330	0.000	0.410	0.565
at $\theta_{NIH} = 1.5$	no	0.188	0.057	3.270	0.001	0.075	0.300
	yes	0.454	0.053	8.520	0.000	0.349	0.558
at $\theta_{NIH} = 2$	no	0.142	0.058	2.430	0.015	0.028	0.257
	yes	0.421	0.067	6.260	0.000	0.289	0.552
at $\theta_{NIH} = 2.5$	no	0.106	0.056	1.910	0.057	−0.003	0.215
	yes	0.388	0.080	4.830	0.000	0.231	0.546

Predictive margins. Model VCE: OIM - SE estimated using Delta-method; n = 600.

Table D1
Marginal effects of the interaction between θ_{NIH} and STD_efforts (Model 8)

Pr (INTENSITY)	at #STD_effort	Margin	Std.Err.	z	P > z	[95%Conf.	Interval]
at $\theta_{NSH} = -2.8$	low	2.438	1.023	2.380	0.017	0.433	4.444
	high	1.261	0.414	3.050	0.002	0.450	2.072
at $\theta_{NSH} = -2.3$	low	1.941	0.670	2.900	0.004	0.628	3.253
	high	1.184	0.322	3.680	0.000	0.553	1.815
at $\theta_{NSH} = -1.8$	low	1.544	0.422	3.660	0.000	0.718	2.371
	high	1.111	0.242	4.590	0.000	0.637	1.585
at $\theta_{NSH} = -1.3$	low	1.229	0.253	4.850	0.000	0.733	1.726
	high	1.043	0.174	5.980	0.000	0.701	1.385
at $\theta_{NSH} = -0.8$	low	0.978	0.148	6.610	0.000	0.688	1.268
	high	0.979	0.122	8.030	0.000	0.740	1.218
at $\theta_{NSH} = -0.3$	low	0.778	0.098	7.960	0.000	0.587	0.970
	high	0.919	0.092	9.960	0.000	0.738	1.099
at $\theta_{NSH} = 0.2$	low	0.620	0.090	6.870	0.000	0.443	0.796
	high	0.862	0.093	9.250	0.000	0.680	1.045
at $\theta_{NSH} = 0.7$	low	0.493	0.097	5.060	0.000	0.302	0.684
	high	0.809	0.115	7.010	0.000	0.583	1.036
at $\theta_{NSH} = -1.2$	low	0.392	0.103	3.800	0.000	0.190	0.595
	high	0.760	0.144	5.270	0.000	0.477	1.042
at $\theta_{NSH} = 1.7$	low	0.312	0.105	2.990	0.003	0.107	0.517
	high	0.713	0.173	4.130	0.000	0.374	1.052
at $\theta_{NSH} = 2.2$	low	0.249	0.102	2.440	0.015	0.049	0.448
	high	0.669	0.199	3.360	0.001	0.279	1.060

Predictive margins. Model VCE: OIM - SE estimated using Delta-method; n = 305.

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